

BABS1201 Course Themes

Course Themes

The course encompasses four major themes. These are not presented in turn, but integrated throughout the course.

Theme 1: Thinking like a scientist.

This theme introduces the skills of scientific thinking, including how to decide what is true or plausible, and how scientists communicate. It also exposes you to cutting edge research being conducted at UNSW. Classes on this theme are interspersed through the term, enabling you:

- To comprehend that science is a never-ending exploration, and that knowledge is provisional.
- To identify the principal characteristics of scientific evidence.
- To understand how scientists approach the investigation of a topic.
- To communicate the principles of scientific findings to other scientists.

Theme 2: Cell biology and cell architecture

This theme describes the principal types of living cells, the key components of cell structure, their functions, and how they relate to each other. Lectures and practical classes on this theme should enable you:

1. To understand the evolutionary origins of life, and of the diversity of life.
2. To identify the different types of living cells, and the main similarities and differences between them.
3. To explain how different cell types are identified.
4. To describe important cell structures and relate these to function.
5. To compare and contrast cell structures in eukaryotes and bacteria.

Theme 3: Metabolism

This theme outlines the key concepts of metabolism, the consumption and generation of energy by living cells. Lectures and practical classes on this theme should enable you:

- To describe the essential differences between proteins, carbohydrates and lipids.
- To describe the processes by which these molecules enter cells.

- To comprehend the processes of generating energy for cellular function.
- To compare and contrast energy generation in animals and plants.

Theme 4: Genetics

This theme introduces the key concepts of modern genetics, including what genes are, how they are regulated, how genetic information is transmitted and how modern molecular biology can use genetics to understand biology. Lectures and practical classes on this theme should enable you:

- To describe the essential structures of genetic material (nucleic acids, genes, chromosomes).
- To explain the processes by which cells divide.
- To describe the principal steps in the control of gene expression and the production of functional proteins.
- To relate these structures and processes to the inheritance of genetic characteristics.
- To explain the uses of recombinant DNA technology in at least one situation relating to investigation of gene function.